

Kilian Shi

Seeking Summer 2027 Internship | Available May 15 – Aug 17

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EDUCATION

Carnegie Mellon University

Expected May 2028

Master of Science in Mobile and IoT Engineering (MSMITE)

Chongqing University

Jul 2026

Bachelor of Engineering in Computer Science and Technology

GPA: 88/100 — TOEFL: 102 — GRE: 331

PROFESSIONAL EXPERIENCE

Ericsson (China) — AI Engineering Intern

Sep 2025 – Nov 2025

- Architected an end-to-end Fusion-RAG search platform in Django, integrating BM25 and ChromaDB with Reciprocal Rank Fusion (RRF) to enhance enterprise query relevance.
- Designed an automated LLM-as-a-Judge evaluation pipeline (Hit Rate, MRR) with dynamically generated QA datasets, driving a ~10% overall improvement in retrieval accuracy.
- Optimized system latency and context window utilization by decoupling retrieval data and developing a HyDE-inspired query reformatting module to accurately map complex user intents.
- Implemented robust full-stack functionalities, including real-time DB syncing and isolated data access boundaries.

Beijing Silu Innovation Technology — Back-End Development Intern

Jun 2025 – Jul 2025

- Migrated legacy enterprise web modules from Spring MVC to a modern Spring Boot and Vue stack.
- Developed a generic web scraping tool and automated documentation generation pipelines for code repositories.

RESEARCH & CORE PROJECTS

U-KAST: Kolmogorov-Arnold Swin Transformer for Medical Segmentation

Aug 2025 – Oct 2025

Accepted by IEEE ISBI 2026 (*arXiv:2511.04084*)

- Developed U-KAST by integrating Group Rational KANs into Swin Transformer encoders to replace standard MLPs, resolving the computational overhead and memory bottlenecks of vanilla spline-based KANs.
- Parameterized edge functions via Safe Pade Activation Units and channel group-sharing, slashing computational cost to 1.43 GFLOPs with a compact 7.2M parameter footprint.
- Overcame the data-hungry nature of ViTs, achieving SOTA accuracy across four diverse 2D/3D medical benchmarks and demonstrating extreme robustness in data-scarce scenarios (using only 10%–25% of training data).

D-GEM: Sparsely Supervised Surgical Video Segmentation

Aug 2025 – Feb 2026

Accepted by MICCAI 2026

- Co-developed a framework utilizing self-supervised DINOv3 representations, decoupling the memory mechanism from the encoder to enable architecture-agnostic and highly data-efficient domain adaptation.
- Engineered a reliability-gated dual-memory system (Transient & Evolving Anchor Memory); formulated gating heuristics and cosine similarity metrics inspired by SAM2/SAM3 temporal propagation mechanics.
- Achieved 74.2% average mIoU across three surgical video benchmarks under 10-frame sparse supervision, outperforming SAM3 by 10.3% while effectively mitigating temporal drift in long-horizon sequences.

SIGN: Sperm Head Morphology Classification with Vision Foundation Models

Aug 2025 – Present

Ongoing Publication Effort

- Architected an INR-guided framework utilizing SAM3 and frozen Foundation Models, replacing heavy dual-stream convolutional branches to drastically reduce parameter complexity.
- Built a SIREN-based DeepSDF auto-decoder to compress 2D masks into 256-D latent codes; injected shape priors into backbone features via Feature-wise Linear Modulation structured by contrastive losses.
- Established SOTA performance on two clinical benchmarks (69.1 F1 on SCIAN-PA; 98.7 F1 on HuSHeM).

Intelligent Recognition System for Dental Lesions (Undergrad Thesis)

Sep 2025 – May 2026

Independent Developer

- Fine-tuned YOLOv8-Seg to extract fine-grained instance masks (0.782 mAP), mitigating cross-tooth noise.
- Designed a Panoramic Context Module and Mask-Guided FiLM within a Swin Transformer backbone to restore global anatomical topology, integrating Rank-MIL pooling to capture micro-lesions in long-tail distributions.
- Outperformed modern baselines (SwinViT, ConvNeXt, etc.) with 0.861 F1/AUC; optimized end-to-end inference speed to 110 FPS and successfully deployed the pipeline as a real-time web diagnostic application.

SKILLS

Languages: Python, Java, C/C++, PostgreSQL

ML & AI: Foundation Models, RAG, NeRF / INR, JAX, Hugging Face

Infra & Tools: PyTorch, Apache Spark, Kafka, Neo4j, ChromaDB, GCP, FastAPI, Docker